Task 1: Create Object Layer

Please find in the git folder name - **OBJECT_LAYER_DESIGN.md**

Path:

https://github.com/sauravbos-lang/caracare_casestudy/blob/main/OBJECT_LAYER_DESIGN.md

Dashboard:

Please find in the git folder name - Prescription Analytics Patient Retention & Marketing Attribution - Casestudy -Dashboard.pdf

Path :

https://github.com/sauravbos-lang/caracare_casestudy/blob/main/Prescription%20Analytics%20Patient%20Retention%20%26%20Marketing%20Attribution%20-%20Casestudy%20-Dashboard.pdf

Task 2: Calculate Represcription Rate

```sql
-- Task 2: Calculate Monthly Re-prescription Rate (Patient-Level)
-- Using Silver Layer: caracare_casestudy.01_silver.fact_prescription
-- Metric: (Distinct patients who got re-prescribed / Total distinct patients) per month

SELECT
  -- Extract the month from prescription_start date
  DATE_TRUNC('MONTH', prescription_start) AS month,
  -- Count of distinct patients with any prescription in the month
  COUNT(DISTINCT patient_id) AS total_patients,
  -- Count of distinct patients who were re-prescribed in the month
  COUNT(DISTINCT CASE WHEN represcription = TRUE THEN patient_id END) AS represcribed_patients,
  -- Calculate re-prescription rate as percentage (represcribed / total), rounded to 2 decimals
  ROUND(
    CAST(COUNT(DISTINCT CASE WHEN represcription = TRUE THEN patient_id END) AS DOUBLE)
    / NULLIF(COUNT(DISTINCT patient_id), 0) * 100,
    2
  ) AS represcription_rate_pct
FROM caracare_casestudy.01_silver.fact_prescription
-- Exclude records with null prescription_start date
WHERE prescription_start IS NOT NULL
-- Group results by month
GROUP BY DATE_TRUNC('MONTH', prescription_start)
-- Order results by month ascending
ORDER BY month
```

Output:

Table

	month	total_patients	represcribed_patients	represcription_rate_pct
1	2025-01-01T00:00:00.000+00:...	2	1	50
2	2025-02-01T00:00:00.000+00:...	2	1	50
3	2025-03-01T00:00:00.000+00:...	1	1	100

Task 3: Touchpoint-Attribution

```sql
-- ============================================================================
-- Task 3: Touchpoint Attribution Analysis
-- Business Question: "Which touchpoint contributes the most to re-prescription?"
-- ============================================================================

-- ANALYTICAL APPROACH: Last-Touch Attribution
-- We assign 100% credit to the LAST touchpoint that occurred during each patient's
-- treatment cycle before they renewed. This is the industry-standard approach for
-- attribution - the final interaction gets credit for "closing" the conversion.

-- METHODOLOGY:
-- 1. Identify all patients who re-prescribed (represcription = TRUE)
-- 2. For each re-prescription, find the LAST touchpoint that occurred during treatment
-- 3. Attribute 100% credit for that re-prescription to that touchpoint type
-- 4. Aggregate to see which touchpoint types contributed most to total re-prescriptions

WITH represcriptions AS (
  -- Step 1: Get all re-prescriptions
  SELECT
    prescription_id,
    patient_id,
    prescription_start,
    prescription_end,
    represcription_date
  FROM caracare_casestudy.01_silver.fact_prescription
  WHERE represcription = TRUE
),

last_touchpoint_per_patient AS (
  -- Step 2: Find the LAST touchpoint during treatment for each re-prescription
  SELECT
    r.prescription_id,
    r.patient_id,
    t.touchpoint_type,
    t.touchpoint_channel,
    t.touchpoint_outcome,
    t.touchpoint_date,
    DATEDIFF(t.touchpoint_date, r.prescription_start) AS days_after_rx_start,
    ROW_NUMBER() OVER (
      PARTITION BY r.prescription_id
      ORDER BY t.touchpoint_date DESC  -- Get the LAST touchpoint
    ) AS recency_rank
  FROM represcriptions r
  INNER JOIN caracare_casestudy.01_silver.fact_touchpoint t
    ON r.prescription_id = t.prescription_id
    AND t.touchpoint_date <= COALESCE(r.represcription_date, r.prescription_end)
  -- Touchpoints must occur during the treatment window
)

-- Step 3: Aggregate attribution credit by touchpoint type
SELECT
  touchpoint_type,
  COUNT(DISTINCT patient_id) AS attributed_represcriptions,
  ROUND(
    CAST(COUNT(DISTINCT patient_id) AS DOUBLE) /
    (SELECT COUNT(DISTINCT patient_id) FROM represcriptions) * 100,
    2
  ) AS attribution_percentage,
  ROUND(AVG(days_after_rx_start), 1) AS avg_days_after_rx_start,
  -- Show breakdown by channel and outcome for context
  CONCAT_WS(', ', COLLECT_SET(CONCAT(touchpoint_channel, ' (', touchpoint_outcome, ')'))) AS channels_and_outcomes
FROM last_touchpoint_per_patient
WHERE recency_rank = 1  -- Only the LAST touchpoint gets credit
GROUP BY touchpoint_type
ORDER BY attributed_represcriptions DESC

-- ============================================================================
-- ✅ ANSWER TO BUSINESS QUESTION:
-- ============================================================================
--
-- CHECK_IN_CALL contributes the most to re-prescription:
--
-- • 2 out of 3 patients (66.67%) who re-prescribed had CHECK_IN_CALL as their
--   last touchpoint
-- • These calls happened on average 65 days after prescription start
-- • All were delivered via PHONE and marked as completed
--
-- EMAIL_NUDGE is secondary:
--
-- • 1 out of 3 patients (33.33%) had EMAIL_NUDGE as their last touchpoint
-- • This email was sent 75 days after prescription start
-- • It was clicked by the patient
--
-- ============================================================================
-- ✅ ANALYTICAL APPROACH:
-- ============================================================================
--
-- Last-Touch Attribution Logic:
--
-- 1. For each patient who re-prescribed, identify the LAST touchpoint that
--    occurred during their treatment
-- 2. Assign 100% credit for that re-prescription to that touchpoint type
-- 3. Aggregate to see which touchpoint types "closed" the most renewals
--
-- Why Last-Touch?
--
```

```
-- • Industry standard for conversion attribution
-- • The final interaction before renewal gets credit for "closing the deal"
-- • Simple, actionable insight: "What was the last thing we did that led to renewal?"
--
-- ============================================================================
-- □□ LIMITATIONS:
-- ============================================================================
--
-- 1. RECENCY BIAS
--    • Gives all credit to the final touchpoint
--    • Ignores earlier touchpoints that may have built awareness, trust, and intent
--
-- 2. NO CAUSALITY PROOF
--    • We see correlation, not causation
--    • Patients may have already decided to renew before the last touchpoint
--
-- 3. SMALL SAMPLE SIZE
--    • Only 3 re-prescriptions total
--    • Results are descriptive, not statistically significant
--
-- 4. MISSING CONTEXT
--    • External factors (doctor relationship, health outcomes, insurance) not captured
--    • No comparison to patients who didn't re-prescribe
--
-- 5. SINGLE-TOUCH BIAS
--    • Assigns 100% credit to one touchpoint, ignoring the entire customer journey
--    • Early touchpoints (awareness, consideration) get no credit
--
-- 6. MISSING TOUCHPOINTS
--    • We only see touchpoints in our system
--    • External factors (word-of-mouth, doctor recommendations, health events)
--      are invisible
--
-- 7. TIME WINDOW ARBITRARY
--    • We use ALL historical touchpoints
--    • A touchpoint from 6 months ago may be less relevant than one from last week
--
-- 8. NO CONTROL GROUP
--    • We don't compare re-prescribed patients to non-re-prescribed patients
--    • Maybe everyone gets these touchpoints, but only some re-prescribe for other reasons
--
-- ============================================================================
-- □ BUSINESS RECOMMENDATION:
-- ============================================================================
--
-- Given that CHECK_IN_CALL accounts for 66.67% of re-prescription attribution:
--
-- HYPOTHESIS: "Personal phone check-ins at ~2 months drive renewal decisions"
--
-- ACTION: Prioritize CHECK_IN_CALL touchpoints for all patients approaching
--         60-70 days into treatment
--
-- NEXT STEPS (with larger dataset):
--
-- • Test: Does adding CHECK_IN_CALL increase retention vs. control?
-- • Optimize: What's the ideal timing? (current avg: 65 days)
-- • Personalize: Which patients benefit most from calls vs. emails?
--
-- BETTER APPROACHES FOR FUTURE ANALYSIS:
--
-- • Multi-touch attribution (linear, time-decay, position-based)
-- • Propensity score matching (compare similar patients with/without touchpoints)
-- • Randomized controlled trials (A/B testing touchpoint strategies)
-- • Machine learning models (feature importance, SHAP values)
-- • Marketing Mix Modeling (econometric approach to isolate touchpoint effects)
--
-- ============================================================================
```

Output:

Table

	touchpoint_type	attributed_represcriptions	attribution_percentage	avg_days_after_rx_start	channels_and_outcomes
1	CHECK_IN_CALL	2	66.67	65	PHONE (completed)
2	EMAIL_NUDGE	1	33.33	75	EMAIL (clicked)